Sourena Khanzadeh

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PROFESSIONAL PROFILE

PhD-trained AI Researcher with experience in conducting research and experiments and publishing papers to advance the state of the art.

TECHNICAL EXPERTISE

Agent systems Prompt engineering, retrieval-augmented generation (RAG), tool calling, multi-agent orchestration, context and memory management, fine-tuning, guardrails, failure analysis, quantization.

ML & languages Python, C/C++, Java; reinforcement learning, heuristic search.

Infrastructure Docker, Git, GitHub Actions, APIs, CI/CD.

PROFESSIONAL EXPERIENCE

AI Researcher, Agentic Systems May 2026 – Present
Flybits Toronto (Full-time)

- Research and develop agentic AI systems, focusing on autonomous reasoning, tool use, memory, and multi-agent orchestration.
- Translate AI research into technical prototypes, research publications, and R&D grant proposals.

AI Researcher, sAIpien May 2026 – Present
MIT Media Lab Part-time Research Affiliation

- Develop perspective-aware AI agents with privacy-preserving memory and adaptive knowledge graphs (Chronicles).
- Build evaluation benchmarks for agent transparency and human oversight, and prototype local AI agents for knowledge sharing and task delegation.

Graduate Researcher Jan 2023 – Apr 2026
Toronto Metropolitan University Toronto

- Designed and evaluated AI/ML methods, producing 7 research publications.

Research Scientist Intern Jan 2024 – Jan 2025
National Research Council Canada Toronto

- Developed and evaluated machine learning approaches for cybersecurity, focusing on intrusion detection and threat identification.

EDUCATION & DISTINCTIONS

Ph.D., Computer Science Sep 2023 – May 2026
Toronto Metropolitan University | **GPA: A+**

Bachelor of Computer Science Sep 2018 – Dec 2022
Toronto Metropolitan University | **Dean's List, 2020–2021**

PUBLICATIONS

- Breadth-First Search vs. Restarting Random Walks for Escaping Uninformed Heuristic Regions.
- An exploratory study on domain knowledge infusion in deep learning for automated threat defense.
- GANsemble for Small and Imbalanced Data Sets: A Baseline for Synthetic Microplastics Data.
- GasZero: A Neuro-Symbolic Approach to Solidity Gas Optimization.
- Solosphere: A framework for gas optimization in solidity smart contracts
- Optimizing gas consumption in ethereum smart contracts: Best practices and techniques
- Integrating Auxiliary Knowledge into Machine Learning to Improve the Detection of Cyberattacks